

4.4.15

AI25BTECH11019 - MENAVATH SAI SANJANA

Question:

Find the sum of the intercepts cut off by the plane $-\mathbf{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) - 5 = 0$ on the three coordinate axes.

Solution:

The given plane is

$$-\mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} - 5 = 0 \implies \mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + 5 = 0 \quad (0.1)$$

X-intercept: $(x, 0, 0)$

$$\mathbf{r} = \begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + 5 = 0 \implies 2x + 5 = 0 \quad (0.2)$$

$$x = -\frac{5}{2} \quad (0.3)$$

Y-intercept: $(0, y, 0)$

$$\mathbf{r} = \begin{pmatrix} 0 \\ y \\ 0 \end{pmatrix}, \quad \mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + 5 = 0 \implies y + 5 = 0 \quad (0.4)$$

$$y = -5 \quad (0.5)$$

Z-intercept: $(0, 0, z)$

$$\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix}, \quad \mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + 5 = 0 \implies -z + 5 = 0 \quad (0.6)$$

$$z = 5 \quad (0.7)$$

Conclusion: The intercepts cut off are

$$x = -\frac{5}{2}, \quad y = -5, \quad z = 5. \quad (0.8)$$

Hence, the sum of intercepts is

$$-\frac{5}{2} - 5 + 5 = -\frac{5}{2}. \quad (0.9)$$

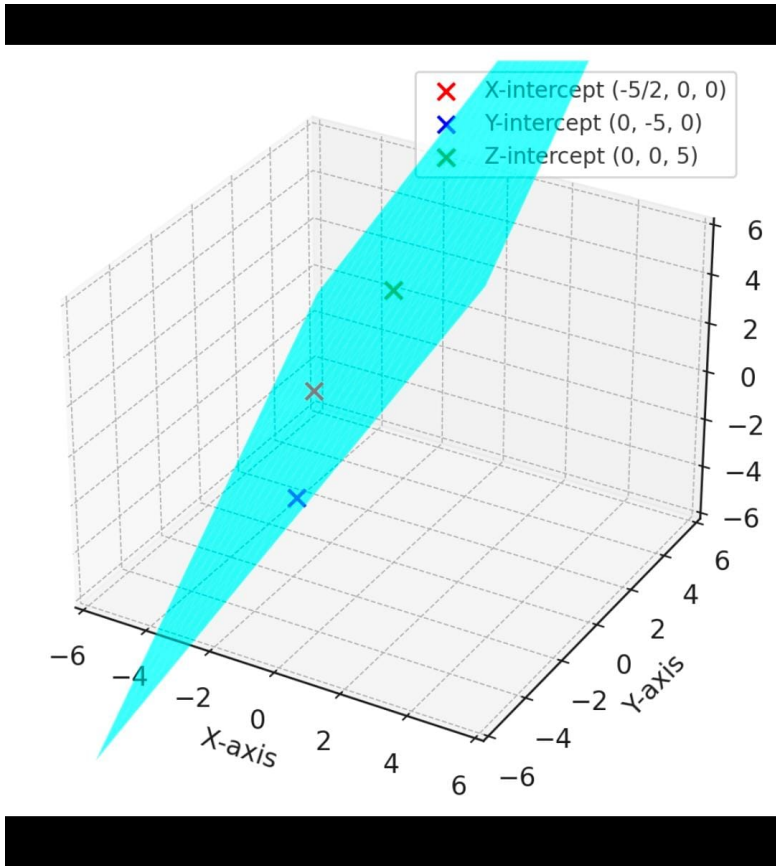


Fig. 0.1